

Radhika Kumbhar

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A dedicated and aspiring Software Engineer with an objective of working in an organization that provides high reputation with a scope of improving knowledge and further career growth.

Education

KITs College of Engineering, Kop

Kolhapur, Maharashtra

Bachelor of Technology in Computer Science

Graduated with distinction (8.5 CGPA)

2021-2025

Experience

Ignnitte Center of Excellence LLP

Aug-present

Working as a Trainer at Ignnitte COE (Center of Excellence) in Skilling and Innovation on the Industry 4.0 module involves teaching, mentoring, and hands-on experience in advanced manufacturing and digital transformation technologies.

Internship

Internship of Deep learning | Edtech Society IITB

2023

Completed Internship at IIT Bombay in deep learning subject is Image Classification with vision Transformer

Projects

Attendance Management System | HTML, CSS, JS, Flask, MySQL [\[Project link\]](#)

- Developed a web-based system for automating employee attendance tracking at Impulse Company.
- Implemented secure login, real-time attendance marking, and database-driven record management.
- Designed a responsive UI and integrated Flask backend with MySQL for efficient data handling.

Image Classification with Vision Transformer (ViT) | Python, Keras, TensorFlow, ViT

- Developed a **Vision Transformer (ViT)** model to perform image classification on the **CIFAR-100** dataset, targeting 100 fine-grained object categories.
- Implemented **multi-head self-attention** and **dense layers** using TensorFlow and Keras to enhance the model's ability to capture spatial relationships and contextual information.
- Performed **hyperparameter tuning** (learning rate, batch size, number of attention heads, etc.) to optimize performance.

SAR Image Colorization with Hybrid cGAN (CNN + ViT) | ViT, CNN, Pytorch

- Designed and implemented a **hybrid conditional Generative Adversarial Network (cGAN)** architecture that combines **Convolutional Neural Networks (CNN)** and **Vision Transformers (ViT)** to colorize grayscale **Synthetic Aperture Radar (SAR)** images.
- Integrated the **CNN encoder** for spatial feature extraction and a **ViT-based decoder** for capturing long-range dependencies and global context during color generation.
- Evaluated model performance using image quality metrics such as **Structural Similarity Index Measure (SSIM)** and **Peak Signal-to-Noise Ratio (PSNR)**.

Medical ChatBot Tool | RAG, Langchain, OpenAI, Flask, Html, CSS [\[github link\]](#)

- Developed an **AI-powered chatbot** capable of answering medical queries by extracting context-aware information from **medical PDFs**.
- Integrated **LangChain** with **Pinecone** vector database and **retrieval-augmented generation**.
- Implemented **document parsing** and **chunk-based querying** to ensure accurate, real-time responses from large unstructured documents.
- Built an intuitive frontend using **Flask, HTML, and CSS**, interactive real-time medical assistance.

Technical Skills

Languages: Python, Java, SQL, HTML, CSS, JavaScript(basic)

Mathematics for ML and DL : Statistics

Python Libraries & Frameworks: Langchain , Langgraph , Pytorch, Pandas , Numpy , Scikit Learn , Pydantic , Streamlit , Flask

Developer Tools: Git, Github, VS Code, Docker, Jupyter Notebook, Kaggle , Google Colab

Data Science: Supervised Machine Learning Algos, Unsupervised Machine Learning Algos, Deep Learning , Exploratory Data Analysis , Feature Engineering , NLP

Achievements

International AR Project Competition| MFU Thailand **2024**
Achieved second place in an AR project competition with the innovative 'Learn ABC' project, developed using Unity.

Eureka Idea Pitching Competition IIT Bombay **2022**
Idea pitching competition which has been held by IITB E-cell. I was the Zonal competitor for that competition – Projectoscope

Soft Skills

Analytical | Communication | Collaborator | Leader | Adaptable | Management | Learner